

第二章 均匀热力学系统的性质

§ 1 Maxwell 关系

$$dU = TdS - PdV$$

$$dH = dU + PdV + VdP = TdS + VdP$$

$$dF = dU - TdS - SdT = -SdT - PdV$$

$$dG = dU - TdS - SdT + PdV + VdP = VdP - SdT$$

$$(1) U = U(S, V)$$

$$T = \left(\frac{\partial U}{\partial S}\right)_V, \quad -P = \left(\frac{\partial U}{\partial V}\right)_S \quad \frac{\partial^2 U}{\partial S \partial V} = \frac{\partial^2 U}{\partial V \partial S}$$

$$\Rightarrow \left(\frac{\partial T}{\partial V}\right)_S = -\left(\frac{\partial P}{\partial S}\right)_V$$

$$(2) H = H(S, P) \rightarrow \left(\frac{\partial T}{\partial P}\right)_S = \left(\frac{\partial V}{\partial S}\right)_P$$

$$(3) F = F(T, V) \rightarrow \left(\frac{\partial S}{\partial V}\right)_T = \left(\frac{\partial P}{\partial T}\right)_V$$

$$(4) G = G(P, T) \rightarrow \left(\frac{\partial V}{\partial T}\right)_P = -\left(\frac{\partial S}{\partial P}\right)_T$$

证法:

$$(a) dU = \left(\frac{\partial U}{\partial T}\right)_V dT + \left(\frac{\partial U}{\partial V}\right)_T dV$$

$$dS = \left(\frac{\partial S}{\partial T}\right)_V dT + \left(\frac{\partial S}{\partial V}\right)_T dV$$

$$dU = TdS - PdV = T \cdot \left(\frac{\partial S}{\partial T}\right)_V dT + \left[T \cdot \left(\frac{\partial S}{\partial V}\right)_T - P\right] dV$$

$$\rightarrow \text{等容热容 } C_V = \left(\frac{\partial U}{\partial T}\right)_V = T \cdot \left(\frac{\partial S}{\partial T}\right)_V$$

$$\rightarrow \left(\frac{\partial U}{\partial V}\right)_T = T \left(\frac{\partial S}{\partial V}\right)_T - P = T \cdot \left(\frac{\partial P}{\partial T}\right)_V - P \quad \rightarrow \text{物态方程导出. } g(P, T, V) = 0$$

例: 证明对理想气体 $\left(\frac{\partial U}{\partial V}\right)_T = 0$, 对范氏气体 $\left(\frac{\partial U}{\partial V}\right)_T = \frac{a}{V^2}$

$$(1) \left(\frac{\partial U}{\partial V}\right)_T = T \cdot \left(\frac{\partial P}{\partial T}\right)_V - P = T \cdot \frac{NR}{V} - P = 0$$

$$(2) \left(P + \frac{a}{V^2}\right)(V - b) = RT \rightarrow P = \frac{RT}{V - b} - \frac{a}{V^2}$$

$$\left(\frac{\partial U}{\partial V}\right)_T = T \cdot \frac{R}{V - b} - P = \frac{a}{V^2}$$

$$(b) dH = \left(\frac{\partial H}{\partial T}\right)_P dT + \left(\frac{\partial H}{\partial P}\right)_T dP = TdS + VdP$$

$$dS = \left(\frac{\partial S}{\partial T}\right)_P dT + \left(\frac{\partial S}{\partial P}\right)_T dP$$

$$dH = T \cdot \left(\frac{\partial S}{\partial T}\right)_P dT + \left[T \left(\frac{\partial S}{\partial P}\right)_T + V\right] dP$$

$$\rightarrow \text{等压热容 } C_P = \left(\frac{\partial H}{\partial T}\right)_P = T \cdot \left(\frac{\partial S}{\partial T}\right)_P \quad \text{物态方程给出}$$

$$\rightarrow \left(\frac{\partial H}{\partial P}\right)_T = T \left(\frac{\partial S}{\partial P}\right)_T + V = V - T \cdot \left(\frac{\partial V}{\partial T}\right)_P$$

$$C_V = \left(\frac{\partial U}{\partial T}\right)_V = T \cdot \left(\frac{\partial S}{\partial T}\right)_V, \quad C_P = \left(\frac{\partial H}{\partial T}\right)_P = T \cdot \left(\frac{\partial S}{\partial T}\right)_P$$

$$C_P - C_V = T \cdot \left[\left(\frac{\partial S}{\partial T}\right)_P - \left(\frac{\partial S}{\partial T}\right)_V \right]$$

$$S = S(T, P) = S(T, V(T, P))$$

$$\left(\frac{\partial S}{\partial T}\right)_P = \left(\frac{\partial S}{\partial T}\right)_V + \left(\frac{\partial S}{\partial V}\right)_T \cdot \left(\frac{\partial V}{\partial T}\right)_P$$

由 $g(P, V, T) = 0$ 给出.

$$C_P - C_V = T \cdot \left(\frac{\partial S}{\partial V}\right)_T \cdot \left(\frac{\partial V}{\partial T}\right)_P = T \cdot \underbrace{\left(\frac{\partial P}{\partial T}\right)_V \cdot \left(\frac{\partial V}{\partial T}\right)_P}$$

对理想气体:

$$C_P - C_V = T \cdot \frac{nR}{V} \cdot \frac{nR}{P} = nR$$

$$\text{膨胀} \sim \alpha = \frac{1}{V} \left(\frac{\partial V}{\partial T}\right)_P, \quad \text{等温压缩} \sim \kappa_T = -\frac{1}{V} \left(\frac{\partial V}{\partial P}\right)_T$$

$$\rightarrow C_P - C_V = T \cdot \frac{\left(\frac{\partial V}{\partial T}\right)_P^2 \cdot \left(\frac{\partial P}{\partial T}\right)_V}{\left(\frac{\partial V}{\partial P}\right)_T} = T \cdot \frac{V^2 \alpha^2}{-\left(\frac{\partial V}{\partial P}\right)_T} = \frac{TV\alpha^2}{\kappa_T}$$

例: 证明 $\frac{\kappa_S}{\kappa_T} = \frac{C_V}{C_P}$, 其中 $\kappa_S = -\frac{1}{V} \left(\frac{\partial V}{\partial P}\right)_S$, $\alpha_T = -\frac{1}{V} \left(\frac{\partial V}{\partial P}\right)_T$

$$\text{证: } \left(\frac{\partial V}{\partial T}\right)_V \cdot \left(\frac{\partial V}{\partial P}\right)_T = \left(\frac{\partial H}{\partial T}\right)_P \cdot \left(\frac{\partial V}{\partial P}\right)_S$$

$$\left(\frac{\partial X}{\partial Y}\right)_Z = \begin{vmatrix} \left(\frac{\partial X}{\partial Y}\right)_V & \left(\frac{\partial Z}{\partial Y}\right)_V \\ \left(\frac{\partial X}{\partial Z}\right)_Y & \left(\frac{\partial Z}{\partial Z}\right)_Y \end{vmatrix} = \frac{\partial(X, Z)}{\partial(Y, Z)}$$

$$T \left(\frac{\partial S}{\partial T}\right)_V \cdot \left(\frac{\partial V}{\partial P}\right)_T = T \cdot \left(\frac{\partial S}{\partial T}\right)_P \cdot \left(\frac{\partial V}{\partial P}\right)_S$$

$$\text{其中 } \left(\frac{\partial Z}{\partial Y}\right)_X \cdot \left(\frac{\partial X}{\partial Z}\right)_Y = \frac{1}{\left(\frac{\partial Y}{\partial X}\right)_Z} = 0$$

$$\frac{\left(\frac{\partial V}{\partial P}\right)_S}{\left(\frac{\partial V}{\partial P}\right)_T} = \frac{\frac{\partial(V, S)}{\partial(P, S)}}{\frac{\partial(V, T)}{\partial(P, T)}} = \frac{\frac{\partial(V, S)}{\partial(V, T)}}{\frac{\partial(P, S)}{\partial(P, T)}} = \frac{\left(\frac{\partial S}{\partial T}\right)_V}{\left(\frac{\partial S}{\partial T}\right)_P} \quad \text{证.}$$

例: 证明 $C_P - C_V = -T \cdot \frac{\left(\frac{\partial P}{\partial T}\right)_V^2}{\left(\frac{\partial V}{\partial T}\right)_T}$

$$S = S(T, P) = S(T, V(T, P))$$

$$\text{LHS} = T \left(\frac{\partial S}{\partial V}\right)_T \cdot \left(\frac{\partial V}{\partial T}\right)_P \quad \left(\frac{\partial S}{\partial V}\right)_T = -\left(\frac{\partial P}{\partial T}\right)_V$$

$$\text{RHS} = -T \left(\frac{\partial P}{\partial T}\right)_V \frac{\frac{\partial(P, V)}{\partial(T, V)}}{\frac{\partial(P, T)}{\partial(V, T)}} = T \cdot \left(\frac{\partial S}{\partial V}\right)_T \left(\frac{\partial V}{\partial T}\right)_P$$

可测量: 状态变量 $(P, V, T, \dots)$

热容, α, β, κ

不可测量: $U, H, F, G, \dots$

§2 节流过程, J-T 效应

节流过程: $\delta Q = 0$, $W = P_1 V_1 - P_2 V_2 = U_2 - U_1$

$$\rightarrow H_1 = H_2$$

定义 J-T 系数: $\mu = \left(\frac{\partial T}{\partial P}\right)_H = \frac{1}{C_p} \cdot [T \cdot \left(\frac{\partial V}{\partial T}\right)_P - V]$

由: $H = H(T, P) = H(S(T, P), P)$

$$\left(\frac{\partial T}{\partial P}\right)_H = - \frac{\left(\frac{\partial H}{\partial P}\right)_T}{\left(\frac{\partial H}{\partial T}\right)_P} = - \frac{1}{C_p} \left[\left(\frac{\partial H}{\partial P}\right)_S + \left(\frac{\partial H}{\partial S}\right)_P \cdot \left(\frac{\partial S}{\partial P}\right)_T \right]$$

$$= - \frac{1}{C_p} \cdot [V - T \cdot \left(\frac{\partial V}{\partial T}\right)_P] = \frac{1}{C_p} [T \cdot \left(\frac{\partial V}{\partial T}\right)_P - V]$$

$$= \frac{V}{C_p} (T\alpha - 1)$$

$$dH = TdS + VdP$$

$$\left(\frac{\partial S}{\partial P}\right)_T = - \left(\frac{\partial V}{\partial T}\right)_P$$

理想气体: $\alpha = \frac{1}{V} \left(\frac{\partial V}{\partial T}\right)_P = \frac{1}{T}$, $\mu = 0$

反转曲线: $\mu = 0$ 的曲线

例: 求范氏气体的反转曲线

$$(P + \frac{N^2 a}{V^2}) (V - Nb) = NRT$$

$$\mu = \frac{1}{C_p} [T \cdot \left(\frac{\partial V}{\partial T}\right)_P - V] = - \frac{1}{C_p} [T \cdot \left(\frac{\partial P}{\partial T}\right)_T \left(\frac{\partial V}{\partial T}\right)_T + V]$$

$$= - \frac{1}{C_p} \cdot \left(\frac{\partial V}{\partial P}\right)_T \cdot [T \left(\frac{\partial P}{\partial T}\right)_T + V \left(\frac{\partial P}{\partial V}\right)_T]$$

$$= - \frac{1}{C_p} \cdot \left(\frac{\partial V}{\partial P}\right)_T \left\{ T \cdot \frac{NR}{V - Nb} + V \cdot \left[- \frac{NR}{(V - Nb)^2} + \frac{2N^2 a}{V^3} \right] \right\}$$

$$= - \frac{1}{C_p} \cdot \left(\frac{\partial V}{\partial P}\right)_T \left[\frac{1}{(V - Nb)^2} (NRTV - NRTV - N^2 bRT) + \frac{2N^2 a}{V^2} \right] = 0$$

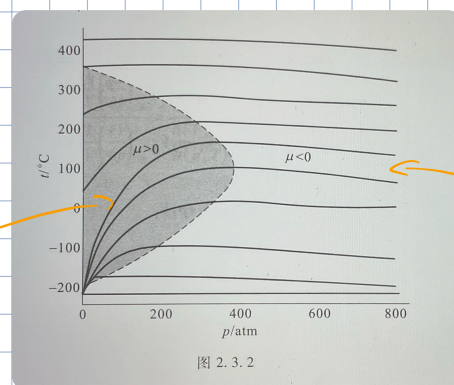
$$\rightarrow \left(1 - \frac{Nb}{V}\right)^2 = \frac{bRT}{2a}$$

解得 $V = \frac{Nb}{1 - \sqrt{\frac{bRT}{2a}}}$

$$P = NRT \cdot \frac{1 - \sqrt{\frac{bRT}{2a}}}{Nb} - \frac{N^2 a}{V^2} = \frac{N^2 a}{V^2} \left(1 - \sqrt{\frac{bRT}{2a}}\right)^2$$

$$= \left(1 - \sqrt{\frac{bRT}{2a}}\right) \left[\frac{RT}{b} \sqrt{\frac{2a}{bRT}} - \frac{a}{b^2} \left(1 - \sqrt{\frac{bRT}{2a}}\right) \right]$$

$$= \frac{a}{b^2} \left(1 - \sqrt{\frac{bRT}{2a}}\right) \left(3 \sqrt{\frac{bRT}{2a}} - 1\right)$$



Onnes 过程:

$$-p_f \cdot p = \frac{nRT}{V} \left(1 + \frac{n}{V} B(T) \right) \approx \frac{nRT}{V} \left(1 + \frac{p}{RT} B(T) \right)$$

$$\hookrightarrow V = \frac{nRT}{p} + nB(T)$$

$$\mu = \frac{1}{C_p} \cdot \left[T \cdot \left(\frac{nR}{p} + n \frac{\partial B}{\partial T} \right) - V \right] = \frac{1}{C_p} \left[T \frac{nR}{V} \frac{\partial B}{\partial T} - nB(T) \right]$$

$$= \frac{n}{C_p} \left[T \frac{\partial B}{\partial T} - B(T) \right]$$

$T \ll T_{\text{env}}$: 吸引显著, $B < 0 \rightarrow \mu > 0$

$T \gg T_{\text{env}}$: 斥力显著, $B > 0 \rightarrow \mu < 0$

§ 3 绝热去磁降温

$$dW = -p dV + \mu_0 H V dM$$

$$dU = T dS - p dV + \mu_0 H V dM$$

$$C_M = \frac{\partial U}{\partial T} = T \left(\frac{\partial S}{\partial T} \right)_M$$

$$C_H = \frac{\partial U}{\partial T} = T \left(\frac{\partial S}{\partial T} \right)_H \quad \left(\frac{\partial T}{\partial H} \right)_S = - \left(\frac{\partial S}{\partial H} \right)_T \left(\frac{\partial T}{\partial S} \right)_H$$

$$\hookrightarrow \left(\frac{\partial T}{\partial H} \right)_S = - \frac{T}{C_H} \left(\frac{\partial S}{\partial H} \right)_T$$

$$\left(\frac{\partial S}{\partial H} \right)_T = \mu_0 \left(\frac{\partial M}{\partial T} \right)_H$$

$$= - \frac{\mu_0 T}{C_H} \left(\frac{\partial M}{\partial T} \right)_H$$

$$\text{构造 } d(TS + \mu_0 H V M - U)$$

$$M = \frac{C_H}{T}, \quad \left(\frac{\partial M}{\partial T} \right)_H = - \frac{C_H}{T^2}$$

$$= S dT - p dV + \mu_0 H V dM + \mu_0 V M dH$$

$$= \frac{\mu_0 C_H}{C_H T} > 0$$

(单位体积)

绝热条件下, 磁场 $\downarrow$, $T \downarrow$ (去磁降温)

§ 4 热辐射的热力学理论

1. 内能密度 u 只与 T 有关

$$u = \frac{U}{V} = u(T)$$

证: ...

$$U = U(V, T) = V \cdot u(T)$$

2. 辐射压强

电磁场的动量张量:

$$P_{xx} = \frac{1}{8\lambda} (\vec{E}_x^2 - \vec{E}_y^2 - \vec{E}_z^2) + \frac{1}{8\lambda} (\vec{H}_x^2 - \vec{H}_y^2 - \vec{H}_z^2)$$

$$P_{xy} = \frac{1}{4\lambda} (\vec{E}_x \vec{E}_y + \vec{H}_x \vec{H}_y)$$

各向同性:

$$\overline{E_x^2} = \overline{E_y^2} = \overline{E_z^2} = \frac{1}{3} \overline{E^2}$$

$$\overline{E_x E_y} = \overline{E_y E_z} = \overline{E_z E_x} = 0$$

$$\Rightarrow P = -P_{xx} = \frac{1}{24\lambda} (\overline{E^2} + \overline{H^2})$$

$$u = \frac{1}{8\lambda} (\overline{E^2} + \overline{H^2}) \quad \text{Gauss 单位制, Poynting 定理给出}$$

$$\hookrightarrow P = \frac{1}{3} u$$

3. 热辐射的热力学函数

$$u(T, V) = V \cdot u(T) \rightarrow \left(\frac{\partial u}{\partial V} \right)_T = u,$$

$$\left(\frac{\partial P}{\partial T} \right)_V = \frac{1}{3} \frac{du}{dT}$$

$$\left(\frac{\partial u}{\partial V} \right)_T = T \cdot \left(\frac{\partial P}{\partial T} \right)_V - P \rightarrow u = T \cdot \frac{1}{3} \frac{du}{dT} - \frac{1}{3} u$$

$$\Rightarrow u = aT^4, \quad P = \frac{1}{3} u = \frac{1}{3} aT^4$$

$$\text{熵: } dS = \frac{du + P dV}{T}$$

$$= \frac{1}{T} d(aT^4 V) + \frac{1}{3} aT^3 dV$$

$$= 4aT^2 V dT + \frac{4}{3} aT^3 dV = \frac{4}{3} a d(VT^3)$$

$$S = \int dS = \frac{4}{3} aT^3 V + S_0 \quad \underline{=0}, S|_{V=0} = 0$$

$$\text{Gibbs 函数: } G = u - TS + PV$$

$$= aT^4 V - T \cdot \frac{4}{3} aT^3 V + \frac{1}{3} aT^4 V = 0$$

光数
不守恒

$$\text{焓、自由能: } H = u + PV = \frac{4}{3} aT^4 V$$

$$F = u - TS = -\frac{1}{3} aT^4 V$$

$$\text{热容: } C_V = \left(\frac{\partial U}{\partial T} \right)_V = 4aT^3 V = 3S$$

$$C_P = T \cdot \left(\frac{\partial S}{\partial T} \right)_P = \infty$$

§ 5. 气体的热力学函数

1. 以 T, V 为参量, 求 u, s

$$P = P(T, V),$$

$$(1) \quad dU = \left(\frac{\partial U}{\partial T}\right)_V dT + \left(\frac{\partial U}{\partial V}\right)_T dV$$

$$C_V = \left(\frac{\partial U}{\partial T}\right)_V = T \left(\frac{\partial S}{\partial T}\right)_V, \quad \left(\frac{\partial U}{\partial V}\right)_T = T \left(\frac{\partial P}{\partial T}\right)_V - P$$

$$dU = C_V dT + [T \left(\frac{\partial P}{\partial T}\right)_V - P] dV$$

$$\rightarrow u = \int [C_V dT + [T \left(\frac{\partial P}{\partial T}\right)_V - P] dV] + u_0$$

$$(2) \quad dS = \left(\frac{\partial S}{\partial V}\right)_T dV + \left(\frac{\partial S}{\partial T}\right)_V dT$$

$$\left(\frac{\partial S}{\partial T}\right)_V = \frac{1}{T} C_V, \quad \left(\frac{\partial S}{\partial V}\right)_T = \left(\frac{\partial P}{\partial T}\right)_V \quad (\text{Maxwell, 3rd})$$

$$dS = \frac{1}{T} C_V dV + \left(\frac{\partial P}{\partial T}\right)_V dT$$

$$S = \int \left[\frac{C_V}{T} dV + \left(\frac{\partial P}{\partial T}\right)_V dT \right] + s_0$$

2. T, P 为参量, 求 u, s

$$V = V(T, P)$$

$$(1) \quad dH = \left(\frac{\partial H}{\partial T}\right)_P dT + \left(\frac{\partial H}{\partial P}\right)_T dP$$

$$C_P = \left(\frac{\partial H}{\partial T}\right)_P = T \left(\frac{\partial S}{\partial T}\right)_P, \quad \left(\frac{\partial H}{\partial P}\right)_T = V - T \left(\frac{\partial V}{\partial T}\right)_P$$

$$dH = C_P dT + [V - T \left(\frac{\partial V}{\partial T}\right)_P] dP$$

$$H = \int [C_P dT + [V - T \left(\frac{\partial V}{\partial T}\right)_P] dP] + H_0$$

$$u = H - P[V(T, P)]$$

$$(2) \quad dS = \left(\frac{\partial S}{\partial T}\right)_P dT + \left(\frac{\partial S}{\partial P}\right)_T dP$$

$$\left(\frac{\partial S}{\partial T}\right)_P = \frac{1}{T} C_P, \quad \left(\frac{\partial S}{\partial P}\right)_T = - \left(\frac{\partial V}{\partial T}\right)_P \quad (\text{M. 4th})$$

$$dS = \frac{C_P}{T} dT - \left(\frac{\partial V}{\partial T}\right)_P dP$$

$$S = \int \left[\frac{C_P}{T} dT - \left(\frac{\partial V}{\partial T}\right)_P dP \right] + s_0$$

$$G = H - TS$$

$$\frac{1}{T} \int C_P dT - \int \frac{1}{T} \cdot C_P dT = - \int \frac{dT}{T^2} \int C_P dT$$

$$= \int [C_P dT + [V - T \left(\frac{\partial V}{\partial T}\right)_P] dP] - T \int \left[\frac{C_P}{T} dT - \left(\frac{\partial V}{\partial T}\right)_P dP \right] + G_0$$

$$(\text{理想气体}) = - T \int \frac{dT}{T^2} \int C_P dT + RT \ln P + G_0 = RT (\varphi + \ln P)$$

$$\varphi = \frac{h_0}{RT} - \frac{C_P \ln T}{R} + \frac{C_P - S_0}{R}$$

例：求范氏气体的内能、熵。

$$(p - \frac{a}{v^2})(v-b) = RT$$

以 v, T 为参量：

$$p(v, T) = \frac{RT}{v-b} + \frac{a}{v^2}, \quad (\frac{\partial p}{\partial T})_v = \frac{R}{v-b}$$

$$\rightarrow U = \int (C_v dT + \frac{a}{v^2} dv) + U_0 = \int C_v dT - \frac{a}{v} + U_0$$

$$S = \int (\frac{C_v}{T} dv + \frac{R}{v-b} dT) + S_0 = \int \frac{C_v}{T} dv + R \ln(v-b) + S_0$$

§ 6 特性函数

某 1 热力学函数 $\rightarrow$ 全部热力学函数, eg. $U(S, P), H(S, P)$

1. $F \rightarrow U, H, G$

$F(T, V), G(T, P)$

$$dF = -SdT - PdV, \quad S = -\frac{\partial F}{\partial T}, \quad P = -\frac{\partial F}{\partial V}$$

$$U = F + TS = F - T \cdot \frac{\partial F}{\partial T} \quad \text{Gibbs-Helmholtz 方程.}$$

$$H = U + PV = F - T \cdot \frac{\partial F}{\partial T} - V \cdot \frac{\partial F}{\partial V}$$

$$G = F + PV = F - V \cdot \frac{\partial F}{\partial V}$$

2. $G \rightarrow U, H, F$

$$dG = -SdT + VdP, \quad S = -(\frac{\partial G}{\partial T})_P, \quad V = (\frac{\partial G}{\partial P})_T$$

$$U = G - PV + TS = G - P \cdot (\frac{\partial G}{\partial P})_T - T \cdot (\frac{\partial G}{\partial T})_P$$

$$H = G + TS = G - T \cdot (\frac{\partial G}{\partial T})_P$$

$$F = G - PV = G - P \cdot (\frac{\partial G}{\partial P})_T$$

例：求表面系统的热力学函数。

物态方程： $f(\phi, A, T) = 0, \quad \phi = \phi(T)$

$$dW = \phi dA$$

$$dF = -SdT + dW = -SdT + \phi dA, \quad S = -\frac{\partial F}{\partial T}$$

$$A \rightarrow 0, \text{ 系统不存在 } \rightarrow S = -A \frac{d\phi}{dT}$$

$$F = \int dF = \int \phi dA = \phi A$$

$$U = F + TS = A(\phi - T \frac{d\phi}{dT})$$

§ 7. 磁介质的热力学函数.

$$dW = \underbrace{V \cdot d\left(\frac{1}{2}\mu_0 H^2\right)}_{\text{激发磁场}} + \underbrace{\mu_0 V H dm}_{\text{介质磁化.}}$$

1. 只有介质, 无磁场

$$dW = \mu_0 V H dm = \mu_0 H dm$$

$$dU = T dS + \mu_0 H dm$$

$$G = U - TS - \mu_0 H M$$

$$dG = -S dT - \mu_0 M dH \rightarrow \left(\frac{\partial G}{\partial T}\right)_H = -S, \left(\frac{\partial G}{\partial H}\right)_T = -\mu_0 M$$

$$\rightarrow \left(\frac{\partial S}{\partial H}\right)_T = \mu_0 \left(\frac{\partial M}{\partial T}\right)_H$$

$$S = S(T, H)$$

$$\rightarrow \left(\frac{\partial S}{\partial T}\right)_H \left(\frac{\partial T}{\partial H}\right)_S \left(\frac{\partial H}{\partial S}\right)_T = -1$$

定义: 热容量 $C_H = T \left(\frac{\partial S}{\partial T}\right)_H$

居里定律: $M = \frac{C_V}{T} H$

$$\begin{aligned} \Rightarrow \left(\frac{\partial T}{\partial H}\right)_S &= -\frac{1}{\left(\frac{\partial S}{\partial T}\right)_H} \cdot \left(\frac{\partial S}{\partial H}\right)_T = -\frac{T}{C_H} \cdot \mu_0 \left(-\frac{C_V}{T^2} H\right) \\ &= \frac{C_V}{C_H T} \mu_0 H \end{aligned}$$

2. 考虑体积变化

$$dU = T dS - p dV + \mu_0 H dm$$

$$G = U - TS + pV - \mu_0 M H$$

$$dG = -S dT + V dP - \mu_0 M dH$$

取 $G = G(P, H)$, $dG = \left(\frac{\partial G}{\partial P}\right)_H dP + \left(\frac{\partial G}{\partial H}\right)_P dH$

$$\Rightarrow \left(\frac{\partial V}{\partial H}\right)_{T, P} = -\mu_0 \left(\frac{\partial M}{\partial P}\right)_{T, H}$$

3. 磁化功

在 x 处, 介质受磁场力 $\mu_0 M(x) \frac{dH(x)}{dx}$

$$W = -\mu_0 \int_{-\infty}^0 M \frac{dH}{dx} dx = -\mu_0 \int_0^{H(0)} M dH$$

$$= -\mu_0 M(0) H(0) + \mu_0 \int_0^{M(0)} H dM$$

$$\rightarrow dW = -\mu_0 M dH$$

$$U^* = -\mu_0 M H + U, \text{ 包含势能项}$$